

TC-01 TO TC-20

Execution Plan Comparison - Complete Test Execution Report

Input - Output - Evidence - Status

Metric	Result
Total test cases	20
Passed	20
Failed	0
Overall result	PASS

Scope: This report covers the Execution Plan Comparison test suite TC-01 through TC-20 executed through the FastAPI Swagger POST /regression-check endpoint.

Evidence basis: Every test below records the observed response fields confirmed during the execution session. Where a screenshot file was available in the session, it is reproduced in the Evidence Appendix. For tests confirmed by exact output but without an available image file, the observed response values are preserved as textual evidence rather than inventing a screenshot.

Final outcome: 20 tests passed, 0 failed. Overall suite status: **PASS**.

Coverage Summary

TC	Scenario	Key proof	Status
TC-01	Same execution plan	plan_changed = false; regression_detected = false; result = No regression.	PASS
TC-02	Index Scan -> Sequential Scan	plan_changed = true; regression_detected = false.	PASS
TC-03	Sequential Scan -> Index Scan	plan_changed = true; improvement_detected = true; regression_detected = false.	PASS
TC-04	Plan change with degradation	plan_changed = true; regression_detected = true; result = Regression detected.	PASS
TC-05	Plan changes but performance improves	plan_changed = true; improvement_detected = true; regression_detected = false.	PASS
TC-06	Estimated cost changes	cost_difference = 15.	PASS
TC-07	Estimated row count changes	rows_difference = 400.	PASS
TC-08	Actual rows change	actual_rows_difference = 250.	PASS
TC-09	JOIN method changes	join_method_changed = true.	PASS
TC-10	JOIN order changes	join_order_changed = true.	PASS
TC-11	Sort operation added	sort_added = true.	PASS

TC	Scenario	Key proof	Status
TC-12	Sort operation removed	sort_removed = true; plan_changed = true.	PASS
TC-13	Filter changes	filter_changed = true.	PASS
TC-14	Scan method changes	scan_method_changed = true.	PASS
TC-15	Plan node added	added_plan_nodes = [SORT].	PASS
TC-16	Plan node removed	removed_plan_nodes = [SORT].	PASS
TC-17	Multiple plan changes	plan_changed = true; join_method_changed = true; join_order_changed = true; sort_removed ...	PASS
TC-18	Same query, different plans	plan_changed = true; regression_detected = false.	PASS
TC-19	Plan comparison report	plan_changed = true; scan_method_changed = true; sort_removed = true; added_plan_nodes = ...	PASS
TC-20	Plan change after index removal	plan_changed = true; scan_method_changed = true; added_plan_nodes = [SEQUENTIAL SCAN]; re...	PASS

Detailed Test Cases

TC-01 - Same execution plan STATUS: PASS

INPUT / TEST ACTION	POST /regression-check; old_query_plan = INDEX SCAN; new_query_plan = INDEX SCAN; old_execution_time = 1; new_execution_time = 1; threshold = 2.
OUTPUT / OBSERVED EVIDENCE	plan_changed = false; regression_detected = false; result = No regression.
VALIDATION DETAIL	Identical old and new execution plans were correctly reported as no plan change.
FINAL STATUS	PASS

TC-02 - Index Scan -> Sequential Scan STATUS: PASS

INPUT / TEST ACTION	old_query_plan = INDEX SCAN; new_query_plan = SEQUENTIAL SCAN; execution times held equal.
OUTPUT / OBSERVED EVIDENCE	plan_changed = true; regression_detected = false.
VALIDATION DETAIL	The engine detected the plan transition without falsely classifying a performance regression.
FINAL STATUS	PASS

TC-03 - Sequential Scan -> Index Scan STATUS: PASS

INPUT / TEST ACTION	old_query_plan = SEQUENTIAL SCAN; new_query_plan = INDEX SCAN; execution time 3 -> 1.
OUTPUT / OBSERVED EVIDENCE	plan_changed = true; improvement_detected = true; regression_detected = false.
VALIDATION DETAIL	A plan change that improved execution time was detected and correctly treated as an improvement.
FINAL STATUS	PASS

Detailed Test Cases (continued)

TC-04 - Plan change with degradation STATUS: PASS

INPUT / TEST ACTION	old_query_plan = INDEX SCAN; new_query_plan = SEQUENTIAL SCAN; execution time 1 -> 3; threshold = 2.
OUTPUT / OBSERVED EVIDENCE	plan_changed = true; regression_detected = true; result = Regression detected.
VALIDATION DETAIL	The system reported both the plan change and the measured performance degradation.
FINAL STATUS	PASS

TC-05 - Plan changes but performance improves STATUS: PASS

INPUT / TEST ACTION	old_query_plan = SEQUENTIAL SCAN; new_query_plan = INDEX SCAN; execution time 5 -> 2.
OUTPUT / OBSERVED EVIDENCE	plan_changed = true; improvement_detected = true; regression_detected = false.
VALIDATION DETAIL	The plan changed, performance improved, and no false regression was raised.
FINAL STATUS	PASS

TC-06 - Estimated cost changes STATUS: PASS

INPUT / TEST ACTION	old_estimated_cost = 10; new_estimated_cost = 25.
OUTPUT / OBSERVED EVIDENCE	cost_difference = 15.
VALIDATION DETAIL	The estimated-cost delta was calculated and returned correctly.
FINAL STATUS	PASS

Detailed Test Cases (continued)

TC-07 - Estimated row count changes STATUS: PASS

INPUT / TEST ACTION	old_estimated_rows = 100; new_estimated_rows = 500.
OUTPUT / OBSERVED EVIDENCE	rows_difference = 400.
VALIDATION DETAIL	The change in estimated rows was identified correctly.
FINAL STATUS	PASS

TC-08 - Actual rows change STATUS: PASS

INPUT / TEST ACTION	old_actual_rows = 100; new_actual_rows = 350.
OUTPUT / OBSERVED EVIDENCE	actual_rows_difference = 250.
VALIDATION DETAIL	The response reported the actual row-count difference correctly.
FINAL STATUS	PASS

TC-09 - JOIN method changes STATUS: PASS

INPUT / TEST ACTION	old_join_method = NESTED LOOP; new_join_method = HASH JOIN.
OUTPUT / OBSERVED EVIDENCE	join_method_changed = true.
VALIDATION DETAIL	The JOIN algorithm change was detected.
FINAL STATUS	PASS

Detailed Test Cases (continued)

TC-10 - JOIN order changes STATUS: PASS

INPUT / TEST ACTION	old_join_order = A -> B -> C; new_join_order = B -> A -> C.
OUTPUT / OBSERVED EVIDENCE	join_order_changed = true.
VALIDATION DETAIL	The change in JOIN order was detected.
FINAL STATUS	PASS

TC-11 - Sort operation added STATUS: PASS

INPUT / TEST ACTION	old_has_sort = false; new_has_sort = true.
OUTPUT / OBSERVED EVIDENCE	sort_added = true.
VALIDATION DETAIL	The newly introduced Sort operation was identified.
FINAL STATUS	PASS

TC-12 - Sort operation removed STATUS: PASS

INPUT / TEST ACTION	old_has_sort = true; new_has_sort = false; old/new plan text differs.
OUTPUT / OBSERVED EVIDENCE	sort_removed = true; plan_changed = true.
VALIDATION DETAIL	The removed Sort operation was identified and the overall plan change was reported.
FINAL STATUS	PASS

Detailed Test Cases (continued)

TC-13 - Filter changes STATUS: PASS

INPUT / TEST ACTION	old_filter = age > 18; new_filter = age > 30.
OUTPUT / OBSERVED EVIDENCE	filter_changed = true.
VALIDATION DETAIL	The filter-condition change was detected.
FINAL STATUS	PASS

TC-14 - Scan method changes STATUS: PASS

INPUT / TEST ACTION	old_scan_method = INDEX SCAN; new_scan_method = BITMAP HEAP SCAN.
OUTPUT / OBSERVED EVIDENCE	scan_method_changed = true.
VALIDATION DETAIL	The scan-method transition was detected.
FINAL STATUS	PASS

TC-15 - Plan node added STATUS: PASS

INPUT / TEST ACTION	old_plan_nodes = [INDEX SCAN]; new_plan_nodes = [INDEX SCAN, SORT].
OUTPUT / OBSERVED EVIDENCE	added_plan_nodes = [SORT].
VALIDATION DETAIL	The additional plan node was identified exactly.
FINAL STATUS	PASS

Detailed Test Cases (continued)

TC-16 - Plan node removed STATUS: PASS

INPUT / TEST ACTION	old_plan_nodes = [INDEX SCAN, SORT]; new_plan_nodes = [INDEX SCAN].
OUTPUT / OBSERVED EVIDENCE	removed_plan_nodes = [SORT].
VALIDATION DETAIL	The removed plan node was identified exactly.
FINAL STATUS	PASS

TC-17 - Multiple plan changes STATUS: PASS

INPUT / TEST ACTION	Old: INDEX SCAN + NESTED LOOP + SORT; New: BITMAP HEAP SCAN + HASH JOIN. JOIN order A -> B changed to B -> A; sort removed; filter age > 18 changed to age > 30; execution time 1 -> 3.
OUTPUT / OBSERVED EVIDENCE	plan_changed = true; join_method_changed = true; join_order_changed = true; sort_removed = true; filter_changed = true; scan_method_changed = true; added_plan_nodes = [BITMAP HEAP SCAN, HASH JOIN]; removed_plan_nodes = [INDEX SCAN, NESTED LOOP, SORT]; regression_detected = true.
VALIDATION DETAIL	All important concurrent plan differences were returned in the same comparison response.
FINAL STATUS	PASS

Detailed Test Cases (continued)

TC-18 - Same query, different plans STATUS: PASS

INPUT / TEST ACTION	historical_query_text = current_query_text = SELECT * FROM users WHERE id = 1;; old plan = INDEX SCAN; new plan = SEQUENTIAL SCAN; execution time 1 -> 1.
OUTPUT / OBSERVED EVIDENCE	plan_changed = true; regression_detected = false.
VALIDATION DETAIL	The same SQL text was compared correctly even though the execution plan differed.
FINAL STATUS	PASS

TC-19 - Plan comparison report STATUS: PASS

INPUT / TEST ACTION	Old plan = INDEX SCAN + SORT; New plan = BITMAP HEAP SCAN; old/new scan methods and plan-node arrays supplied.
OUTPUT / OBSERVED EVIDENCE	plan_changed = true; scan_method_changed = true; sort_removed = true; added_plan_nodes = [BITMAP HEAP SCAN]; removed_plan_nodes = [INDEX SCAN, SORT].
VALIDATION DETAIL	The response displayed old/new plan information and the material differences required by the comparison report.
FINAL STATUS	PASS

Detailed Test Cases (continued)

TC-20 - Plan change after index removal STATUS: PASS

INPUT / TEST ACTION	Old plan/method/node = INDEX SCAN; New plan/method/node = SEQUENTIAL SCAN; execution time 1 -> 3; threshold = 2.
OUTPUT / OBSERVED EVIDENCE	plan_changed = true; scan_method_changed = true; added_plan_nodes = [SEQUENTIAL SCAN]; removed_plan_nodes = [INDEX SCAN].
VALIDATION DETAIL	The plan difference produced after index removal was detected correctly.
FINAL STATUS	PASS

Evidence Appendix

The following screenshots are the image evidence files available from this execution session. They are reproduced without alteration except scaling to fit the page.

Server response

Code	Details
200	Response body<pre>"join_method_changed": false, "old_join_order": "", "new_join_order": "", "join_order_changed": false, "old_has_sort": false, "new_has_sort": false, "sort_added": false, "sort_removed": false, "old_filter": "", "new_filter": "", "filter_changed": false, "old_scan_method": "", "new_scan_method": "", "scan_method_changed": false, "added_plan_nodes": ["SORT"], "regression_cause": "Performance degradation", "old_execution_time": 1, "new_execution_time": 1, "slowdown_percent": 0, "regression_threshold": 2, "performance_ratio": 1</pre>

Evidence A - TC-15 response excerpt showing added_plan_nodes = ["SORT"] and the corresponding execution result.

Response body

```
{
  "query_name": "PLAN_TC17",
  "old_query_plan": "INDEX SCAN + NESTED LOOP + SORT",
  "new_query_plan": "BITMAP HEAP SCAN + HASH JOIN",
  "plan_changed": true,
  "old_estimated_cost": 0,
  "new_estimated_cost": 0,
  "cost_difference": 0,
  "old_estimated_rows": 0,
  "new_estimated_rows": 0,
  "rows_difference": 0,
  "old_actual_rows": 0,
  "new_actual_rows": 0,
  "actual_rows_difference": 0,
  "old_join_method": "NESTED LOOP",
  "new_join_method": "HASH JOIN",
  "join_method_changed": true,
  "old_join_order": "A -> B",
  "new_join_order": "B -> A",
  "join_order_changed": true,
  "old_has_sort": true,
  "new_has_sort": false,
  "sort_added": false,
  "sort_removed": true,
  "old_filter": "age > 18",
  "new_filter": "age > 30",
  "filter_changed": true,
}
```

Evidence B1 - TC-17 response: plan_changed, JOIN method/order, sort, filter and related fields.

Response body

```

  "sort_added": false,
  "sort_removed": true,
  "old_filter": "age > 18",
  "new_filter": "age > 30",
  "filter_changed": true,
  "old_scan_method": "INDEX SCAN",
  "new_scan_method": "BITMAP HEAP SCAN",
  "scan_method_changed": true,
  "added_plan_nodes": [
    "BITMAP HEAP SCAN",
    "HASH JOIN"
  ],
  "removed_plan_nodes": [
    "INDEX SCAN",
    "NESTED LOOP",
    "SORT"
  ],
  "regression_cause": "Performance degradation",
  "old_execution_time": 1,
  "new_execution_time": 3,
  "slowdown_percent": 200,
  "regression_threshold": 2,
  "performance_ratio": 3,
  "regression_detected": true,
  "improvement_detected": false,
  "result": "Regression detected"
}
```

Response headers

Evidence B2 - TC-17 response continuation: scan_method_changed, added/removed plan nodes, performance ratio and regression result.

Final Verification Statement

All twenty Execution Plan Comparison test cases, TC-01 through TC-20, produced the expected comparison behavior recorded in this report. No test case remained failed or pending at the end of the session.

Metric	Result
Total test cases	20
Passed	20
Failed	0
Overall result	PASS

Overall Result: PASS - 20/20

Execution Plan Comparison

Edge Case Test Execution Report - EC-01 to EC-25

Complete final status with executed evidence | Report date: 20 Aug 2026

Total Edge Cases	Passed	Failed (Final)	Final Pass Rate
25	25	0	100%

Final result: EC-01 through EC-25 are all PASS after required fixes and retests. EC-17 and EC-21 initially failed, received minimum targeted fixes, and then passed on retest.

Evidence basis: Swagger response values, explicit user confirmations of matching output, full response payloads supplied for EC-17/EC-24/EC-25, and terminal concurrency evidence for EC-19.

Important scope note: EC-20 was executed as a simulated unavailable-plan condition (empty current plan), not by physically shutting down PostgreSQL. EC-21 was executed through the application permission_denied test flag.

Final Status Matrix

ID	Edge Case	Key Final Evidence	Status
EC-01	No old plan	plan_comparison_available=false; plan_changed=null; result="No baseline plan"	PASS
EC-02	No new plan	plan_comparison_available=false; plan_changed=null; result="Current plan unavailable"	PASS
EC-03	Both plans identical	plan_changed=false; regression_detected=false; result="No regression"	PASS
EC-04	Empty plan	plan_comparison_available=false; plan_changed=null; result="Invalid or missing plan"	PASS
EC-05	Malformed plan	HTTP 422 Validation Error; no HTTP 500 and no application crash	PASS
EC-06	Plan node added	plan_changed=true; added_plan_nodes=["SORT"]	PASS
EC-07	Plan node removed	plan_changed=true; removed_plan_nodes=["SORT"]	PASS
EC-08	Node order changed	plan_changed=true	PASS
EC-09	Join method changed	join_method_changed=true; plan_changed=true	PASS
EC-10	Scan method changed	scan_method_changed=true; plan_changed=true	PASS
EC-11	Cost changes only	plan_changed=false; cost_difference=15	PASS
EC-12	Estimated rows changes only	plan_changed=false; rows_difference=400	PASS
EC-13	Actual time changes but plan same	plan_changed=false; regression_detected=true; result="Regression detected"	PASS
EC-14	Plan changes but performance improves	plan_changed=true; improvement_detected=true; regression_detected=false	PASS
EC-15	Very large plan	HTTP 200; plan_changed=false; regression_detected=false	PASS

ID	Edge Case	Key Final Evidence	Status
EC-16	Multiple changes	plan_changed=true; join_method_changed=true; sort_removed=true; scan_method_changed=true; added=["BITMAP HEAP SCAN", "HASH JOIN"]; removed=["INDEX SCAN", "NESTED LOOP", "SORT"]	PASS
EC-17	Same SQL but different constants	After fix/retest: plan_changed=false; regression_detected=false; result="No regression"; no different-query early return	PASS
EC-18	Different SQL queries	query_changed=true; regression_detected=null; result="Different query/version - regression comparison not applied"	PASS
EC-19	Concurrent executions	[(EC19_Q1, False, No regression), (EC19_Q2, True, Regression detected), (EC19_Q3, False, No regression)]	PASS
EC-20	PostgreSQL unavailable	plan_comparison_available=false; plan_changed=null; result="Current plan unavailable"	PASS
EC-21	Permission denied	After control-flow fix/retest: plan_comparison_available=false; plan_changed=null; result="Permission denied while obtaining execution plan"	PASS
EC-22	Plan contains new PostgreSQL node type	HTTP 200; plan_changed=true; added_plan_nodes=["CUSTOM NEW NODE"]; removed_plan_nodes=["INDEX SCAN"]	PASS
EC-23	Statistics changed	plan_changed=true; rows_difference=900	PASS
EC-24	Index created	plan_changed=true; scan_method_changed=true; added=["INDEX SCAN"]; removed=["SEQUENTIAL SCAN"]; improvement_detected=true; regression_detected=false; result="No regression"	PASS
EC-25	Index dropped	plan_changed=true; scan_method_changed=true; added=["SEQUENTIAL SCAN"]; removed=["INDEX SCAN"]; regression_detected=true; result="Regression detected"	PASS

Detailed Execution Evidence

Each case below records the test intent, the actual input condition used, the key observed response, and the evidence trail captured during execution.

EC-01 - No old plan **PASS**

Situation	Historical execution plan does not exist
Expected	Cannot compare; show No baseline plan
Executed Input	old_query_plan=""; new_query_plan="INDEX SCAN"
Observed Output	plan_comparison_available=false; plan_changed=null; result="No baseline plan"
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-02 - No new plan **PASS**

Situation	Current plan unavailable
Expected	Comparison should fail gracefully
Executed Input	old_query_plan="INDEX SCAN"; new_query_plan=""
Observed Output	plan_comparison_available=false; plan_changed=null; result="Current plan unavailable"
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-03 - Both plans identical **PASS**

Situation	Old = New
Expected	Report no plan change
Executed Input	old_query_plan="INDEX SCAN"; new_query_plan="INDEX SCAN"; old_time=1; new_time=1
Observed Output	plan_changed=false; regression_detected=false; result="No regression"
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-04 - Empty plan **PASS**

Situation	Plan data is empty
Expected	Show invalid/missing plan error
Executed Input	old_query_plan=""; new_query_plan=""
Observed Output	plan_comparison_available=false; plan_changed=null; result="Invalid or missing plan"
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-05 - Malformed plan **PASS**

Situation	Plan format is invalid
Expected	Reject/handle safely
Executed Input	old_query_plan supplied as an object instead of a string
Observed Output	HTTP 422 Validation Error; no HTTP 500 and no application crash
Evidence	Swagger returned 422; user explicitly reported "422 came".
Final Status	PASS

EC-06 - Plan node added **PASS**

Situation	New plan has extra node
Expected	Report added node
Executed Input	old_nodes=["INDEX SCAN"]; new_nodes=["INDEX SCAN","SORT"]
Observed Output	plan_changed=true; added_plan_nodes=["SORT"]
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-07 - Plan node removed **PASS**

Situation	Old plan has node missing in new plan
Expected	Report removed node
Executed Input	old_nodes=["INDEX SCAN","SORT"]; new_nodes=["INDEX SCAN"]
Observed Output	plan_changed=true; removed_plan_nodes=["SORT"]
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-08 - Node order changed **PASS**

Situation	Same nodes but different order
Expected	Detect structural difference
Executed Input	old_plan="INDEX SCAN -> SORT -> HASH JOIN"; new_plan="HASH JOIN -> INDEX SCAN -> SORT"
Observed Output	plan_changed=true
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-09 - Join method changed **PASS**

Situation	Nested Loop -> Hash Join
Expected	Detect join change
Executed Input	old_join_method="NESTED LOOP"; new_join_method="HASH JOIN"
Observed Output	join_method_changed=true; plan_changed=true
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-10 - Scan method changed **PASS**

Situation	Index Scan -> Sequential Scan
Expected	Detect scan change
Executed Input	old_scan_method="INDEX SCAN"; new_scan_method="SEQUENTIAL SCAN"
Observed Output	scan_method_changed=true; plan_changed=true
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-11 - Cost changes only **PASS**

Situation	Plan structure same but cost differs
Expected	Report cost difference
Executed Input	same plan; old_estimated_cost=10; new_estimated_cost=25
Observed Output	plan_changed=false; cost_difference=15
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-12 - Estimated rows changes only **PASS**

Situation	Plan structure same, estimates differ
Expected	Report estimate difference
Executed Input	same plan; old_estimated_rows=100; new_estimated_rows=500
Observed Output	plan_changed=false; rows_difference=400
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-13 - Actual time changes but plan same **PASS**

Situation	Same plan, slower execution
Expected	Report performance change without falsely claiming plan change
Executed Input	same plan; old_execution_time=1; new_execution_time=3; threshold=2
Observed Output	plan_changed=false; regression_detected=true; result="Regression detected"
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-14 - Plan changes but performance improves **PASS**

Situation	New plan is faster
Expected	Report change, not regression
Executed Input	old_plan="SEQUENTIAL SCAN"; new_plan="INDEX SCAN"; 5 -> 2
Observed Output	plan_changed=true; improvement_detected=true; regression_detected=false
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-15 - Very large plan **PASS**

Situation	Query produces deeply nested/large plan
Expected	System should compare without crashing
Executed Input	8-node plan: SEQ SCAN, FILTER, SORT, NESTED LOOP, HASH JOIN, AGGREGATE, MATERIALIZE, LIMIT
Observed Output	HTTP 200; plan_changed=false; regression_detected=false
Evidence	Swagger response matched expected output; application did not crash (user confirmed same output).
Final Status	PASS

EC-16 - Multiple changes **PASS**

Situation	Many nodes change
Expected	Report all significant differences
Executed Input	INDEX SCAN + NESTED LOOP + SORT -> BITMAP HEAP SCAN + HASH JOIN; scan/join/sort/nodes changed
Observed Output	plan_changed=true; join_method_changed=true; sort_removed=true; scan_method_changed=true; added=["BITMAP HEAP SCAN","HASH JOIN"]; removed=["INDEX SCAN","NESTED LOOP","SORT"]
Evidence	Swagger response matched all expected flags and node lists (user confirmed same output).
Final Status	PASS

EC-17 - Same SQL but different constants **PASS**

Situation	Different parameter values
Expected	Query identification should follow defined fingerprinting rules
Executed Input	SELECT * FROM users WHERE id = 1; vs SELECT * FROM users WHERE id = 2;
Observed Output	After fix/retest: plan_changed=false; regression_detected=false; result="No regression"; no different-query early return
Evidence	Initial run failed classification. Added constant-normalizing fingerprint logic using regex; retest output was supplied in full and passed.
Final Status	PASS

EC-18 - Different SQL queries **PASS**

Situation	Completely different queries
Expected	Do not compare as the same query
Executed Input	users query vs orders query
Observed Output	query_changed=true; regression_detected=null; result="Different query/version - regression comparison not applied"
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-19 - Concurrent executions **PASS**

Situation	Multiple plan captures simultaneously
Expected	Keep plan records associated with correct query/execution
Executed Input	3 concurrent POST requests via ThreadPoolExecutor(max_workers=3)
Observed Output	[(EC19_Q1, False, No regression), (EC19_Q2, True, Regression detected), (EC19_Q3, False, No regression)]
Evidence	Terminal screenshot captured concurrent results with correct query_name/result association.
Final Status	PASS

EC-20 - PostgreSQL unavailable **PASS**

Situation	Cannot obtain plan
Expected	Show database/collection error
Executed Input	Simulated unavailable current plan: old_query_plan="INDEX SCAN"; new_query_plan=""
Observed Output	plan_comparison_available=false; plan_changed=null; result="Current plan unavailable"
Evidence	Swagger response matched graceful unavailability behavior (user confirmed same output). Note: test simulated unavailable plan rather than physically stopping PostgreSQL.
Final Status	PASS

EC-21 - Permission denied **PASS**

Situation	User cannot obtain required plan information
Expected	Show permission error
Executed Input	permission_denied=true with empty plan fields
Observed Output	After control-flow fix/retest: plan_comparison_available=false; plan_changed=null; result="Permission denied while obtaining execution plan"
Evidence	Initial run returned Invalid or missing plan (FAIL). Permission check was moved before empty-plan validation; retest matched expected output (user confirmed same output).
Final Status	PASS

EC-22 - Plan contains new PostgreSQL node type **PASS**

Situation	Unsupported node
Expected	Handle unknown node gracefully
Executed Input	old_nodes=["INDEX SCAN"]; new_nodes=["CUSTOM NEW NODE"]
Observed Output	HTTP 200; plan_changed=true; added_plan_nodes=["CUSTOM NEW NODE"]; removed_plan_nodes=["INDEX SCAN"]
Evidence	Swagger response matched expected output; unknown node handled without crash (user confirmed same output).
Final Status	PASS

EC-23 - Statistics changed **PASS**

Situation	Same query gets a different plan after ANALYZE/statistics changes
Expected	Detect plan difference
Executed Input	same SQL; INDEX SCAN -> SEQUENTIAL SCAN; estimated rows 100 -> 1000
Observed Output	plan_changed=true; rows_difference=900
Evidence	Swagger response matched expected output (user confirmed same output).
Final Status	PASS

EC-24 - Index created **PASS**

Situation	New index changes plan
Expected	Detect plan change
Executed Input	SEQUENTIAL SCAN -> INDEX SCAN; time 3 -> 1
Observed Output	plan_changed=true; scan_method_changed=true; added=["INDEX SCAN"]; removed=["SEQUENTIAL SCAN"]; improvement_detected=true; regression_detected=false; result="No regression"
Evidence	Full Swagger response was supplied in the conversation and matches all expected values.
Final Status	PASS

EC-25 - Index dropped **PASS**

Situation	Dropping index changes plan
Expected	Detect plan change
Executed Input	INDEX SCAN -> SEQUENTIAL SCAN; time 1 -> 3
Observed Output	plan_changed=true; scan_method_changed=true; added=["SEQUENTIAL SCAN"]; removed=["INDEX SCAN"]; regression_detected=true; result="Regression detected"
Evidence	Full Swagger response was supplied in the conversation and matches all expected values.
Final Status	PASS

Defect and Retest Trail

Case	Initial Result	Minimum Fix	Retest Result
EC-17	FAIL - different constants were classified as a different query/version.	Added regex-based query fingerprints that normalize numeric/string constants before query identity comparison.	PASS - same SQL structure with id=1 vs id=2 proceeded normally; plan_changed=false, regression_detected=false, result=No regression.
EC-21	FAIL - empty-plan validation returned Invalid or missing plan before permission handling.	Moved permission_denied handling before empty-plan validation.	PASS - result=Permission denied while obtaining execution plan.

Evidence Appendix

The following images are included from the conversation as source/evidence artifacts.

Source test specification - EC-01 to EC-13

EC ID	Edge Case	Situation	Expected Behaviour
EC-01	No old plan	Historical execution plan doesn't exist	Cannot compare; show "No baseline plan"
EC-02	No new plan	Current plan unavailable	Comparison should fail gracefully
EC-03	Both plans identical	Old = New	Report no plan change
EC-04	Empty plan	Plan data is empty	Show invalid/missing plan error
EC-05	Malformed plan	Plan format is invalid	Reject/handle safely
EC-06	Plan node added	New plan has extra node	Report added node
EC-07	Plan node removed	Old plan has node missing in new plan	Report removed node
EC-08	Node order changed	Same nodes but different order	Detect structural difference
EC-09	Join method changed	Nested Loop → Hash Join	Detect join change
EC-10	Scan method changed	Index Scan → Seq Scan	Detect scan change
EC-11	Cost changes only	Plan structure same but cost differs	Report cost difference
EC-12	Estimated rows changes only	Plan structure same, estimates differ	Report estimate difference
EC-13	Actual time changes but plan same	Same plan, slower execution	Report performance change without falsely claiming plan change

Source test specification - EC-13 to EC-25

	Changes only	Unchanged	
EC-13	Actual time changes but plan same	Same plan, slower execution	Report performance change without falsely claiming plan change
EC-14	Plan changes but performance improves	New plan is faster	Report change, not regression
EC-15	Very large plan	Query produces deeply nested/large plan	System should compare without crashing
EC-16	Multiple changes	Many nodes change	Report all significant differences
EC-17	Same SQL but different constants	Different parameter values	Query identification should follow defined fingerprinting rules
EC-18	Different SQL queries	Completely different queries	Do not compare as the same query
EC-19	Concurrent executions	Multiple plan captures simultaneously	Keep plan records associated with correct query/execution
EC-20	PostgreSQL unavailable	Cannot obtain plan	Show database/collection error
EC-21	Permission denied	User cannot obtain required plan information	Show permission error
EC-22	Plan contains new PostgreSQL node type	Unsupported node	Handle unknown node gracefully
EC-23	Statistics changed	Same query gets a different plan after ANALYZE/statistics changes	Detect plan difference
EC-24	Index created	New index changes plan	Detect plan change
EC-25	Index dropped	Dropping index changes plan	Detect plan change

EC-19 concurrent execution terminal evidence

```

PS C:\Users\reddy\OneDrive\Desktop\DATA BASE\QUERY Performance Project> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned)
; (& "c:\Users\reddy\OneDrive\Desktop\DATA BASE\QUERY Performance Project\.venv\Scripts\Activate.ps1")
(.venv) PS C:\Users\reddy\OneDrive\Desktop\DATA BASE\QUERY Performance Project> python
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
Ctrl click to launch VS Code Native REPL
>>> from concurrent.futures import ThreadPoolExecutor; import requests
>>> payloads = [{"query_name": "EC19_Q1", "old_query_plan": "INDEX SCAN", "new_query_plan": "INDEX SCAN", "old_execution_time": 1, "new_execution_time": 1, "regression_threshold": 2}, {"query_name": "EC19_Q2", "old_query_plan": "INDEX SCAN", "new_query_plan": "SEQ SCAN", "old_execution_time": 1, "new_execution_time": 3, "regression_threshold": 2}, {"query_name": "EC19_Q3", "old_query_plan": "SEQ SCAN", "new_query_plan": "INDEX SCAN", "old_execution_time": 3, "new_execution_time": 1, "regression_threshold": 2}]
>>> ex=ThreadPoolExecutor(max_workers=3); results=list(ex.map(lambda p: requests.post("http://127.0.0.1:8000/regression-check", json=p).json(), payloads)); ex.shutdown(); print([(r["query_name"], r["regression_detected"], r["result"]) for r in results])
[('EC19_Q1', False, 'No regression'), ('EC19_Q2', True, 'Regression detected'), ('EC19_Q3', False, 'No regression')]
>>>

```

EC-21 initial failure evidence before ordering fix

Code	Details
200	Response body <pre>{ "query_name": "EC_21", "plan_comparison_available": false, "plan_changed": null, "result": "Invalid or missing plan" }</pre> Response headers <pre>content-length: 111 content-type: application/json date: Thu, 20 Aug 2026 05:15:50 GMT server: uvicorn</pre>
Responses	

Final Verification Statement

Execution Plan Comparison Edge Case suite EC-01 through EC-25 is complete. Every listed edge case has a final PASS status. No edge case remains pending in this suite.

Range	Total	Pass	Fail	Final Status
EC-01 to EC-25	25	25	0	COMPLETE - 100% PASS